

Recursive Renormalization and Projective Boundaries: A Proof-Based BRCT Formulation

A substrate-independent derivation of inverse-Zeno depth, projective Bindu limits, nested phase dynamics, and invariant-preserving reconstruction

Martin Timms

Independent Researcher, Electrozest, Northumberland, UK

Research working paper - 25 September 2026

Purpose. The argument is developed as a proof chain. Each derived result is followed by its interpretation, applicability, and limitations. Established results from fluid/gravity correspondence, twistor theory, FIT/GML, and fractal-time-crystal work are cited inline as external inputs rather than folded into the internal proof.

Abstract

We formulate BRCT as a substrate-independent theory of recursive renormalisation and projective boundaries. Starting only from a recursive state space equipped with an external scale, we derive geometric contraction, finite external accumulation with unbounded recursive depth, the inverse-Zeno logarithmic coordinate, and a projective boundary obtained from compatible renormalised states. We define this boundary as Bindu. We then prove that a fixed-point Bindu has no non-trivial internal clock, whereas periodic and quasiperiodic invariant sets induce phase coordinates. Stable relations between those phases define a nested-clock Bindu. Finally, quotienting extended states by a declared invariant algebra yields an invariant-preserving compression/projection theorem: the outward reconstruction need not recover the microscopic state, only an equivalent representative.

The abstract proof chain is then instantiated rather than redefined. The Navier-Stokes vorticity equation yields an exact decomposition of enstrophy growth into stretching, Fisher-information-like amplitude gradients, directional gradients, and forcing. A scale family gives the inverse-square Fisher scaling used to define concentration depth. The known near-horizon fluid/gravity correspondence provides an external bridge from incompressible Navier-Stokes data to vacuum Einstein geometry [1-3]. Penrose/twistor theory provides a second established example in which spacetime fields are encoded by projective data [4,5]. FIT/GML and the 2026 fractal-time-crystal model motivate the nested phase interpretation [6-8].

The framework does not derive quantum gravity or consciousness. Instead, it isolates the missing theorem: additional non-arbitrary boundary axioms must generate quantum structure and gravitational dynamics rather than having them inserted. Recursive AI is proposed as a directly controllable substrate on which the contraction, invariant, phase, and reconstruction hypotheses can be experimentally tested.

1. Primitive structure

Definition 1. A recursive renormalisation system is a tuple $B = (X, R, s)$, where X is a state space, $R: X \rightarrow X$ is a recursive map, and $s: X \rightarrow \mathbb{R}_{\geq 0}$ is an external scale functional.

$$X_{n+1} = R(X_n), \quad s_n = s(X_n).$$

Assumption A1. There exists $0 < r < 1$ such that $s_{n+1} \leq r s_n$ along the orbit of interest.

Lemma 1. Geometric scale contraction

Lemma 1. Under A1, $s_n \leq s_0 r^n$, hence $s_n \rightarrow 0$.

Proof. Iterating A1 gives $s_{n+1} \leq r s_n \leq \dots \leq s_0 r^{n+1}$. Since $0 < r < 1$, the geometric factor tends to zero.

Corollary 1. Finite accumulated external scale

Corollary 1. If $\Delta a_n \leq C s_n$, then infinitely many recursive levels can occur inside finite accumulated external extent.

Proof.

$$\sum_{n=0}^{\infty} \Delta a_n \leq C s_0 \sum_{n=0}^{\infty} r^n = C s_0 / (1-r) < \infty.$$

Why this matters. This is the first substrate-independent result. It applies whenever an external scale contracts multiplicatively: a spatial core, a remaining time interval, a hierarchy of resonator periods, graph resolution, or an AI representation budget. It tells us that 'infinite recursion at a point' need not mean infinite physical size, storage, or energy. It means that the internal recursion index can remain unbounded while the external measure converges.

2. Continuous limit and inverse-Zeno depth

For a trajectory approaching a finite accumulation time T , write $s=T-t$. We seek a coordinate in which multiplicative contraction becomes additive translation. Taking the logarithm gives

$$\tau = -\log(T-t).$$

Theorem 1. Finite external time, infinite internal depth

Theorem 1. The map above sends $t \in [t_0, T)$ to an unbounded depth interval. Equal increments $\Delta\tau$ correspond to geometrically shrinking physical intervals.

Proof. Let $\tau_n = \tau_0 + n\Delta\tau$. Inversion gives

$$T-t_n = \exp(-\tau_0)\exp(-n\Delta\tau),$$

$$t_{n+1}-t_n = \exp(-\tau_0)(1-\exp(-\Delta\tau))\exp(-n\Delta\tau).$$

The intervals form a convergent geometric series whose sum is $T-t_0$. Hence finite external duration contains unbounded logarithmic depth.

Why this matters. The logarithmic coordinate is more than a convenient change of variables. It supplies a common internal time for systems whose physical clocks accelerate as a boundary is approached. Recurrence that looks increasingly rapid in ordinary time may become stationary, periodic, or quasiperiodic in τ . This is the coordinate in which a nested-clock hypothesis can be tested rather than asserted.

If a characteristic scale satisfies $s(\tau) = s_0 \exp(-\beta\tau)$, then

$$d \log s / d\tau = -\beta,$$

so exponential physical contraction becomes uniform translation in recursive depth.

3. Renormalised states and the projective Bindu

Scale contraction alone could end in a featureless point. To retain structure, remove the shrinking scale by a normalisation N_n :

$$Y_n = N_n X_n.$$

A2. $\{Y_n\}$ is precompact. A3. Coarse-graining maps $\pi_{n+1,n}$ make successive renormalised descriptions asymptotically compatible.

Theorem 2. Projective boundary

Theorem 2. Under A2, a convergent renormalised subsequence exists. Under A3, a compatible convergent subsequence defines an element of the inverse/projective limit.

Proof. Precompactness gives $Y_{n_k} \rightarrow Y^*$ after passage to a subsequence. Compatibility makes the limiting representatives consistent under $\pi_{n+1,n}$. Therefore

$$B \in \limleftarrow (Y_n, \pi_{n+1,n}).$$

The compatible limit B is retained while external scale vanishes.

Definition 2 (Bindu). The non-empty projective boundary element or invariant projective set produced by Theorem 2 is called Bindu.

Why this matters. This changes the physical interpretation. Bindu is not required to be a microscopic object containing an impossible amount of literal data. It is a boundary representation of mutually compatible descriptions across scale. The construction is applicable to renormalisation flows, hierarchical dynamical systems, multiscale resonators, compressed computational states, and potentially boundary formulations of spacetime. The important experimental question becomes whether renormalised structure survives as scale contracts.

4. Recurrence derives internal time

Let F be the induced map on renormalised states.

Lemma 2. Fixed points are not clocks

Lemma 2. If $Y_n \rightarrow Y^*$ and $F(Y^*)=Y^*$, any continuous phase observable depending only on Y becomes constant.

Proof. Continuity gives $\phi(Y_n) \rightarrow \phi(Y^*)$.

Theorem 3. Periodicity induces a recursive clock

Theorem 3. If the invariant set is a cycle of minimal period $p>1$, a phase coordinate exists whose advance per recursive step is $2\pi/p$.

Proof. Label the cycle states by

$$\phi(Y_j^*) = 2\pi j/p \mod 2\pi.$$

Since F advances one cycle state, $\phi(F(Y)) - \phi(Y) = 2\pi/p \mod 2\pi$.

A torus rotation $\phi_{n+1} = \phi_n + \Omega \mod 2\pi$ generalises the construction to several quasiperiodic clocks.

Why this matters. This result explains why 'frequency' should not be a primitive BRCT quantity. Physical hertz belongs to a substrate. Recurrence and phase survive a change of substrate. A microtubule may express phase through electromagnetic resonance; a renormalised fluid through recurrence in logarithmic time; an AI system through update cycles. This is the point at which Anirban Bandyopadhyay's nested-clock/FIT-GML programme becomes mathematically comparable without assuming that the underlying physical mechanisms are identical [6-8].

5. From phase to resonance

For two persistent phases define

$$L_{ij}(n) = \text{wrap}[\varphi_i(n) - m_{ij}\varphi_j(n) - \delta_{ij}],$$

$$q_{ij} = |\lim_{N \rightarrow \infty} (1/N) \sum_{n=1}^N \exp(iL_{ij}(n))|.$$

Proposition 1. A projective Bindu is a nested-clock Bindu if its invariant dynamics contains at least two persistent phase coordinates and at least one non-trivial asymptotic phase-lock relation.

Proof. Theorem 3 supplies persistent internal clocks. Non-zero q_{ij} supplies a stable relation between clocks. Hence the boundary contains relational temporal structure rather than recurrence alone.

Why this matters. This gives a falsifiable definition of resonance. A collection of spectral peaks is insufficient. The phases must remain related under increasing depth, perturbation, and reasonable changes of decomposition. In FIT/GML terms, this distinguishes a mere hierarchy from a hierarchy carrying a reproducible geometric musical language. The 2026 fractal-time-crystal paper reports a nested hierarchy spanning Hz through THz; BRCT abstracts from those dimensional bands to the invariant phase relations themselves [8].

6. The two-breath information-processing theorem

The preceding sections define a recursively compressed boundary and the phase structure that may survive there. We now make the information-processing interpretation explicit. The inward breath is a family of compression maps C_n that removes externally redundant degrees of freedom while preserving a declared invariant family J . The outward breath is a family of expansion maps P_n that reconstructs an extended representative from the compressed boundary state.

$$X \rightarrow C_1(X) \rightarrow \dots \rightarrow B \rightarrow \dots \rightarrow P_1(B) = \tilde{X}.$$

The term fractal domain will mean the renormalised projective hierarchy near maximal compression: external scale has contracted, but compatible internal structure, recurrent phase, and invariant relations remain available across recursive depth. It is not assumed to be a literal geometric fractal in every substrate.

Definition 4. Boundary computation

Definition 4. Let B be the projective boundary and let $K:B \rightarrow B$ be a computation performed on the compressed state. The full processing cycle is $P \circ K \circ C$. The highest-compression domain is computationally non-trivial when K changes task-relevant invariants or relations while remaining inside the admissible boundary state space.

Thus compression is not merely a prelude to storage. It can create the domain in which computation is performed. An extended state may contain many degrees of freedom that are irrelevant to the requested transformation. After quotienting them away, K acts directly on the smaller state that represents the relevant equivalence class.

Theorem 4. Invariant-preserving two-breath computation

Theorem 4. Let M be an extended state space and $J:M \rightarrow V$ a declared invariant representation. Suppose $C:M \rightarrow B$ factors through the quotient $M/\sim_J$ and is injective on the quotient. Suppose further that a boundary operation $K:B \rightarrow B$ induces a well-defined transformation $k:V \rightarrow V$. Then there exists an expansion P such that

$$J(P(K(C(X)))) = k(J(X)).$$

Proof. Because C is injective on quotient classes, each compressed state identifies one and only one J -equivalence class. By hypothesis K maps that boundary state to another admissible boundary state corresponding to the transformed invariant value $k(J(X))$. Choose P to return any extended representative of that target equivalence class. By construction its invariant representation is exactly $k(J(X))$.

Why this matters. This proves why processing at maximal compression can matter. If the task depends only on J , then computation need not manipulate the full microscopic state. It can operate on the quotient representation that contains exactly the distinctions relevant to the task. Expansion is postponed until a macroscopic representative is required. In information-processing terms, the expensive dimensionality of the world is separated from the lower-dimensional relational structure on which the transformation acts.

Corollary 4. Computational advantage of the fractal domain

Corollary 4. If the cost of evaluating a task operator grows monotonically with representation size, and if the compressed boundary has strictly smaller effective dimension than the extended state while preserving task sufficiency, then performing the task as $P \circ K \circ C$ requires a smaller task-domain

representation than applying an equivalent operator directly in M.

Proof. Theorem 4 establishes task equivalence on J. By assumption the boundary representation has smaller effective dimension. Therefore the task operator K acts on fewer task-relevant degrees of freedom than a direct operator on the full representation. The theorem does not guarantee lower wall-clock cost because compression and expansion themselves have costs; it proves the representational advantage and identifies the condition under which an end-to-end computational advantage can occur.

This distinction is important. BRCT does not claim that compression automatically makes every computation faster. It claims that when a task is invariant-sufficient at the boundary, the central computation can be carried out in a reduced domain. Whether the complete cycle wins depends on the costs of C, K, and P, and on whether the boundary can be reused across many operations.

Theorem 5. Continuous simultaneous inward and outward flow

The breathing picture need not alternate globally between a completed inhale and a completed exhale. Let $b(\tau)$ denote the compressed boundary state. Let $u_{in}(\tau)$ be information arriving from the extended domain and $u_{out}(\tau)$ the information returned from the boundary. A continuous model is

$$db/d\tau = F_B(b) + G_{in}u_{in} - G_{out}u_{out}$$

$$y_{out} = P(b), \quad u_{in} = C(x_{in}, b).$$

Theorem 5. If the boundary evolution is well posed and C and P are causal maps, non-zero inward and outward information fluxes may coexist over the same interval. No theorem in the two-breath construction requires temporal alternation of the two directions.

Proof. The state equation contains simultaneous input and output couplings. Causality requires only that outputs at depth-time τ depend on present or past boundary state, not on future input. Therefore $u_{in}(\tau)$ and $u_{out}(\tau)$ may both be non-zero while $b(\tau)$ continues to evolve. The sequential composition $P \circ K \circ C$ is recovered as a special pulsed or staged case; continuous bidirectional operation is the more general dynamical system.

Why this matters. This turns the metaphor of two breaths into a duplex computational architecture. Compression and expansion can be continuously interleaved: new information is being folded into the high-compression fractal domain while processed invariant structure is simultaneously being projected outward. A brain, a recurrent AI system, a resonant physical network, or a boundary field need not stop receiving information in order to emit a response. The centre acts less like a buffer that fills and empties and more like an active transform through which counter-propagating information streams are coupled.

Proposition 2. Why simultaneous flow can preserve a stable centre

Proposition 2. Suppose the autonomous boundary dynamics F_B has a locally stable invariant set A, and the net bidirectional forcing remains bounded within its basin of attraction. Then continuous inward/outward flux need not destroy the compressed boundary; the state can remain close to A while processing ongoing perturbations.

Proof sketch. Local stability supplies a neighbourhood in which perturbations decay or remain bounded. Bounded input/output coupling acts as a perturbation of the autonomous boundary dynamics. Standard input-to-state stability reasoning then permits a bounded neighbourhood of A to remain invariant when the forcing is sufficiently small relative to the restoring dynamics. A substrate-specific proof requires the corresponding stability estimates.

Why this matters. This matters because a useful Bindu cannot be only an instantaneous terminal state. For continuous cognition or computation it must be dynamically maintained while information crosses it in both directions. The strongest BRCT implementation is therefore not compress-then-expand once, but a maintained high-compression attractor that continuously absorbs, transforms, and projects information.

Theorem 6. Non-equilibrium steady Bindu with simultaneous counter-flow

To make simultaneity fundamental rather than optional, resolve the exchange into two non-negative information fluxes, J_{in} and J_{out} . They are not the same information travelling in reverse. The inward flux carries newly resolved distinctions toward the compressed domain; the outward flux carries transformed invariant structure toward extended representations.

$$dB/d\tau = F_B(B) + J_{in}(\tau) - J_{out}(\tau).$$

Theorem 6. A stationary or statistically stationary Bindu is compatible with continuous computation when $J_{in} > 0$ and $J_{out} > 0$ simultaneously. In particular, if the mean balance condition $\langle F_B + J_{in} - J_{out} \rangle = 0$ holds while both mean flux magnitudes remain non-zero, the mean boundary state can remain stationary despite persistent information throughput.

Proof. Stationarity constrains the net rate of change of the boundary state, not each constituent flux. Setting the time-averaged derivative to zero gives $\langle F_B \rangle + \langle J_{in} \rangle - \langle J_{out} \rangle = 0$. This equation permits $\langle J_{in} \rangle > 0$ and $\langle J_{out} \rangle > 0$. Therefore zero mean accumulation does not imply zero information exchange. The boundary may be dynamically maintained while information continuously enters and leaves.

Why this matters. This is the stronger form of the two-breath concept. Bindu should not be imagined as a container that first fills during an inhale and then empties during an exhale. It can be a non-equilibrium processing boundary: inward compression and outward expansion are active at the same time, while the compressed state remains statistically stable. A flame, vortex, biological cell, recurrent network, or streaming AI process can maintain recognisable macroscopic organisation despite continual throughput; BRCT asks whether a recursively compressed invariant boundary can do the analogous informational job.

Corollary 6. Processing without boundary accumulation

Corollary 6. If a boundary operator K continuously transforms incoming invariant content before it contributes to J_{out} , then a stationary Bindu may perform non-zero computation even when its stored boundary information does not grow.

Proof. Theorem 6 allows non-zero throughput at zero mean accumulation. If K changes the invariant relation between input and output, then the outgoing flux is not merely a copy of the incoming flux. Hence a transformation has occurred despite stationary boundary occupancy.

Why this matters. This matters computationally because it separates state from throughput. The highest-compression domain need not grow indefinitely as more information is processed. It can behave as a continuously reused transform. For recursive AI this suggests architectures in which a compact invariant state is maintained while context is continuously compressed into it and predictions, actions, retrieval requests, or reconstructed explanations are continuously expanded from it.

The sequential cycle is therefore recovered only when the supports of J_{in} and J_{out} are deliberately separated in time. The general BRCT formulation permits their supports to overlap completely. In that sense the two breaths are better understood as two simultaneously available directions of information flow through recursive scale.

The resulting computational picture is

extended state $\rightleftharpoons$ recursive compression hierarchy $\rightleftharpoons$ fractal/projective boundary
[K] $\rightleftharpoons$ recursive expansion hierarchy $\rightleftharpoons$ extended state,

with both arrow directions active concurrently. The word 'breath' describes the dual direction of information flow, not a requirement for alternating phases.

7. Navier-Stokes: deriving a concrete compression observable

For incompressible flow, with vorticity $\omega = \nabla \times u$, curl forcing $g = \nabla \times f$, and strain S ,

$$D_t \omega = S\omega + \nu \Delta \omega + g.$$

Multiplying by ω and integrating gives the enstrophy identity

$$dZ/dt = 2 \int \omega \cdot S \omega dx - 2\nu \int |\nabla \omega|^2 dx + 2 \int \omega \cdot g dx.$$

Write $\omega = A\xi$. Orthogonality of a unit vector to its derivative gives

$$|\nabla \omega|^2 = |\nabla A|^2 + A^2 |\nabla \xi|^2.$$

With $\rho = A^2/Z$, substitution yields the exact decomposition

$$d \log Z / dt = 2E_\rho [\xi \cdot S \xi] - (\nu/2) I_A - 2\nu I_D + F_\omega,$$

$$I_A = \int \rho |\nabla \log \rho|^2 dx, \quad I_D = \int \rho |\nabla \xi|^2 dx.$$

For $\rho_\ell(x) = \ell^{-d} \rho_0(x/\ell)$, direct change of variables gives

$$I_A[\rho_\ell] = \ell^{-2} I_A[\rho_0].$$

Why this matters. The relevance is not that Fisher information 'proves' Bindu. It provides an exact localisation observable whose scaling can reveal a contracting physical length. If a trajectory has finite-time concentration, Theorem 1 supplies its depth coordinate; the renormalised flow can then be tested for the projective and phase conditions of Theorems 2-3. This converts BRCT from an analogy about singularity into a measurement programme on a concrete nonlinear field theory. Classical regularity work on vorticity direction also emphasises that geometry of the vorticity field matters, not magnitude alone [9].

8. Established bridge I: fluid dynamics to gravity

The gravity step is not derived from BRCT. It enters through an external result. Bredberg, Keeler, Lysov and Strominger explicitly associate incompressible Navier-Stokes solutions in $p+1$ dimensions with vacuum Einstein solutions in $p+2$ dimensions in a controlled near-horizon/hydrodynamic expansion [1]. Related derivations connect Einstein constraints and null-surface dynamics with Navier-Stokes-type equations [2,3].

External Result G1. In its specified regime, incompressible Navier-Stokes boundary data determine a corresponding vacuum Einstein geometry [1].

BRCT fluid orbit $\rightarrow$ renormalised boundary data $\rightarrow_{G1}$ Einstein geometry.

Why this matters. This is why beginning with Navier-Stokes is potentially useful even after the substrate is removed. It gives a calibration case in which a lower-dimensional dynamical description is already known to encode gravitational geometry. The research task is to discover which BRCT invariants survive that correspondence and whether the projective Bindu has a natural interpretation in the gravitational variables. That is much stronger than saying that fluids and spacetime merely look similar.

9. Established bridge II: projective data to spacetime

Twistor theory supplies a separate projection precedent. In the twistor approach spacetime is secondary, with events represented by compact holomorphic curves in twistor space; field equations can be encoded in twistor cohomology [4,5]. A twistor may be written

$$Z^\alpha = (\omega^A, \pi_{A'}), \quad \omega^A = i\chi^{AA'} \pi_{A'}.$$

External Result P1. In the sectors covered by the Penrose transform, suitable projective/twistor data encode and reconstruct spacetime fields [4,5].

Penrose's conformal methods likewise permit an unbounded physical coordinate to be represented at a finite conformal boundary:

$$\hat{g}_{ab} = \Omega^{-2} g_{ab}.$$

Why this matters. The relevance to BRCT is structural but important. Theorem 4 says that a compact boundary representation can be sufficient for reconstruction of selected invariants; twistor theory supplies an established physical example in which projective data encode spacetime fields. Conformal compactification separately demonstrates that 'infinity' can be a finite boundary in another representation. Together these suggest a precise future question: can the BRCT projective limit be expressed in variables on which a Penrose-like reconstruction is mathematically natural?

10. Quantum gravity: deriving the missing obligations

Lemma 3. The classical BRCT axioms do not imply quantum mechanics.

Proof. The axioms admit ordinary deterministic classical dynamical systems as models. They contain no non-commuting observable algebra, complex probability amplitudes, Planck constant, Born rule, or quantum measure. Therefore quantum mechanics is not true in every model of the axioms and cannot be derived from them.

Why this matters. This negative result matters because it identifies the exact gap. Adding quantum terminology to a recursive boundary would not close it. A quantum BRCT must add or derive a principle that forces complex amplitudes and non-commutative observables. Only after that step is there a meaningful question of quantum gravity.

A candidate quantum extension would need at least structures of the form

$$|\psi\rangle = \sum_a c_a |B_a\rangle, \quad [\hat{A}, B] \neq 0,$$

while the gravitational side must recover, rather than assume, an appropriate classical limit such as

$$S_{EH} = (1/16\pi G) \int (R - 2\Lambda) \sqrt{-g} d^4x.$$

Main open theorem. Find additional non-arbitrary axioms on the projective Bindu boundary that uniquely generate, or strongly constrain, both the quantum state structure and a gravitational continuum limit.

Why this matters. If this theorem can be proved, the chain from recursive boundary dynamics to quantum gravity would become a derivation. If it cannot, BRCT may still be useful as a cross-substrate theory of recursive information and reconstruction, but it would not be a fundamental theory of gravity.

11. Penrose reduction as a later consistency test

Penrose's gravity-related reduction proposal assigns a gravitational self-energy to sufficiently distinct mass distributions in superposition and a characteristic timescale of order $\hbar/E_G$. It is a proposed physical mechanism, not an established consequence of BRCT.

$$E_G \sim (G/2) \iint \delta\rho(x)\delta\rho(y)/|x-y| d^3x d^3y, \quad \tau_G \sim \hbar/E_G.$$

Why this matters. Its correct place in the proof chain is after a quantum Bindu state has been defined and projected into mass-energy geometry. Then it supplies a possible coherence constraint on superposed projected geometries. Placing it earlier would make the argument circular: the geometry whose difference enters E_G must first have been generated by the theory.

12. Recursive AI as an experimental realisation

Let H_n be the state of a recursive AI system, including context, retrieval, memory, intermediate representation, and tool results:

$$H_{n+1} = C_{\theta}(H_n, E_n).$$

Choose an effective scale $s(H)$ and pre-register an invariant family J : factual constraints, proof obligations, causal relationships, source commitments, or goal state. If representation scale contracts, Lemma 1 applies. If aligned states are precompact and compatible, Theorem 2 supplies a computational Bindu. Recurrence supplies phase through Theorem 3; boundary sufficiency supplies reconstruction through Theorem 4.

Corollary 5. A recursive AI system is a BRCT realisation for J when it satisfies the contraction, projective-boundary, and invariant-reconstruction hypotheses; it becomes a nested-clock realisation when recurrent phase and phase locking are also observed.

Why this matters. This is immediately applicable. It predicts that repeated summarisation alone should often fail because it discards quotient-relevant invariants. A better architecture explicitly learns or enforces the invariant algebra while allowing irrelevant detail to collapse. Retrieval, memory, planning, verification, and generation can be treated as coupled clocks. The theory therefore suggests measurable design variables for small/local AI: invariant retention, effective dimension, recurrence spectrum, phase coordination, and reconstruction fidelity - rather than simply model size or recursion count.

13. Proof graph, applications and logical status

Step	Derived result	What it enables	Status
A	External contraction gives vanishing scale.	Finite nested hierarchy without infinite external extent.	Internal proof: Lemma 1.
B	Finite accumulation gives unbounded logarithmic depth.	A common internal coordinate for accelerating recursion.	Internal proof: Theorem 1.
C	Compatible precompact renormalisation gives projective Bindu.	Boundary representation independent of vanishing physical scale.	Internal proof under A2-A3: Theorem 2.
D	Recurrent invariant dynamics gives phase.	Substrate-free clocks and resonance tests.	Internal proof: Theorem 3.
E	Phase locking gives nested-clock Bindu.	Operational FIT/GML comparison.	Definition + Proposition 1.
F	Boundary separation of invariant classes gives reconstruction.	Two-breath compression/expansion without microscopic reversibility.	Internal proof: Theorem 4.
G	NS decomposition yields localisation and directional terms.	Concrete field-theory observables for contraction/geometry.	Exact derivation; compare [9].
H	NS data map to Einstein geometry in near-horizon sector.	Calibration of boundary dynamics against classical gravity.	External result [1-3].
I	Twistor data reconstruct spacetime fields.	Precedent for projective outward reconstruction.	External result [4,5].
J	Classical BRCT does not imply quantum mechanics.	Identifies missing quantum axiom rather than hiding it.	Internal negative result: Lemma 3.
K	Boundary axioms should generate gravitational action.	Would turn compatibility into a gravity derivation.	Open theorem.
L	BRCT may be tested in recursive AI.	Direct experimental implementation of A-F.	Proposed experiment.

The wrapped table is intentionally a map of the argument rather than a compressed list of equations. Steps A-F are the new formal BRCT core. G is a concrete realisation. H-I are independently established bridges. J marks a proved limitation. K is the decisive fundamental-physics problem, and L is the most accessible experimental programme.

14. Applicability and interpretation

14.1 Multiscale physical systems

The framework applies wherever a meaningful external scale contracts while a renormalised description remains bounded. Candidate domains include singular or near-singular PDE dynamics, critical phenomena, hierarchical resonators, and other systems studied by renormalisation. The theory adds a specific question to ordinary scaling analysis: does the limit retain an invariant algebra and recurrent phase structure sufficient for reconstruction?

14.2 Resonant supramolecular systems

FIT/GML and fractal-time-crystal work provide a natural empirical setting for the nested-clock part of the theory [6-8]. BRCT does not require the reported biological frequency bands to be universal. It predicts that the stronger evidence would be scale-normalised phase relationships that remain coherent across levels and that predict geometry or state transitions.

14.3 Gravity and quantum geometry

The fluid/gravity and twistor results show that boundary and projective descriptions are already serious tools in gravitational physics [1-5]. BRCT's contribution is to ask whether such boundaries can themselves be derived from a generic recursive-renormalisation process. Its current value is therefore architectural: it identifies a route and the theorem that would have to succeed. It does not yet replace general relativity or any existing quantum-gravity programme.

14.4 Recursive artificial intelligence

AI provides a particularly clean test because representation scale and invariants can be chosen and measured. A system can be forced through hundreds of recursive compress/reconstruct cycles while effective rank, semantic constraints, phase recurrence, and reconstruction error are recorded. Unlike the fundamental-physics case, failure can be observed directly and the architecture changed. This makes AI a useful experimental mathematics platform for the substrate-independent part of BRCT.

14.5 Consciousness

Nothing proved here entails consciousness. A later consciousness hypothesis could propose that phenomenology correlates with a stable, recursively maintained boundary that integrates distributed invariants and reconstructs a coherent world model. That would require independent neuroscientific evidence and controls. Keeping this claim outside the proof chain strengthens rather than weakens the mathematical paper.

15. Conclusion

The paper begins with one assumption - recursive contraction of an external scale - and derives progressively stronger structures only when additional hypotheses justify them. Contraction gives finite accumulation; finite accumulation gives inverse-Zeno depth; compatible renormalisation gives a projective boundary; recurrence gives phase; phase locking gives nested clocks; and separation of invariant equivalence classes gives a rigorous two-breath reconstruction.

$$\textit{contraction} \Rightarrow \textit{depth} \Rightarrow \textit{projective boundary} \Rightarrow \textit{phase} \Rightarrow \textit{resonance} \Rightarrow \textit{reconstruction}.$$

The importance of this order is methodological. No physical substrate is built into the core. Navier-Stokes, supramolecular resonance, gravity, and recursive AI become tests or realisations of a mathematical structure rather than evidence being blended into a single metaphor. Established literature enters exactly where the proof requires an external bridge, with inline citations.

The next mathematical objective is consequently narrow: determine whether additional principles on the projective boundary can uniquely constrain a quantum state structure and gravitational action. Until then, BRCT is a new recursive-boundary framework with concrete applications and falsifiers, not a completed theory of quantum gravity.

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Internal theorem numbering denotes results proved from the stated BRCT assumptions. Citations identify external mathematical or empirical inputs. A proof within the formal system does not establish that a physical system satisfies its assumptions.